

Julian Canales

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I work the messy part between a base model and a shipped agent: post-training, tools, evals.

Experience

Avathon | AI Engineer

Oct 2025 – Present

- Built the knowledge-graph agent behind the Barrick Gold engagement, working directly with client leadership to shape what they valued; it anchored a multi-year enterprise contract now in production.
- Extracted the company's conversational AI from a product monolith into a standalone multi-tenant agent platform (Django async ASGI, LangGraph, LiteLLM gateway) and shipped it through dev, UAT and production for a Fortune-500 deployment; it is now the company's shared agent layer.
- Replaced config-trusted persona scoping with authorization enforced against live identity (Keycloak/OIDC), clearing the last blocker to general availability; added drop-in MCP server support with per-server isolation and forwarded user tokens, taking new tool servers live by configuration with no code and no deploy; first server shipped with 65 tools.
- Designed and materialized the canonical ontology behind the agent's answers, modeling 28.7M rows of client data into an agent-queryable knowledge graph, plus the operations layer that verifies freshness and correctness and runs dependency-ordered recompute, replacing hand-run refresh scripts.
- Own agent quality with no dedicated QA function on the team: built the evaluation and observability stack (Langfuse, per-query cost/latency/token telemetry) and run continuous evals scoring answer correctness and tool-call trajectory across 16 identity-gated personas, catching wrong answers a green test suite misses.
- Fine-tuned Qwen2.5-14B (QLoRA) for natural-language-to-graph-query translation on 1,232 execution-verified pairs, reaching parity with Claude Sonnet on single-query generation under a judge-graded agentic eval at a fraction of per-query API cost.
- Built the carrier scoring and tendering engine for the supply-chain product, separating metric computation from decision policy so scoring is config-driven; drove adoption of DBOS over Celery+Redis, adding crash recovery and audit trails.
- Rebuilt and shipped avathon.com in two weeks (Next.js, Sanity CMS) with the VPs of Strategic Sales and Marketing and a peer engineer, relaunching the company's public face for enterprise buyers and company-wide recruiting.

IBM | Data Science Intern

May – Aug 2023

- Built and deployed a Flask proof-of-concept integrating Watson Assistant with Db2, used in the pitch that won the client engagement.
- Shipped a Dash application on watsonx and Watson Discovery with a custom evaluation API for side-by-side LLM comparison.

UT Austin | Undergraduate Researcher, Earthquake Modeling

Jan – May 2024

- Re-engineered a legacy Mathematica seismic-response solver into a multithreaded Python pipeline, processing 29k+ NGA-West2 records on 64-vCPU GCP sweeps and cutting runtime from ≈ 3 years to ≈ 2 weeks.

Selected Projects

Pane | *Go, CLI, local-first*

2026

- Built and shipped a shared-memory layer for concurrent AI coding agents (Go, distributed via Homebrew tap): session state, intents, inter-agent messaging, file-overlap detection and git guardrails, provider-agnostic at the shell layer, so parallel agents hand off to each other instead of routing context through the human.

Crabs AI Assistant | *OpenClaw, LightRAG, CouchDB*

2026

- Self-hosted personal agent on a Hetzner VPS with CouchDB syncing an Obsidian vault across devices, running two-tier memory (short-term conversational recall plus long-term knowledge-graph traversal via LightRAG) under a zero-guessing retrieval policy that grounds every answer in indexed notes rather than pre-trained weights.

Adaptive DQN Planner | *RL, DQN, CEM, Python*

2025

- Built an RL agent for safe autonomous driving in highway-env, pairing DQN with CEM planning and interval-based robust control, using Bayesian uncertainty and an online k-NN error tracker to tighten bounds under noisy dynamics; reached 0% collisions and +17% safety margin over deterministic baselines.

Education

The University of Texas at Austin — M.S. Artificial Intelligence

Aug 2025 – May 2027

Machine Learning, Deep Learning, Advances in Deep Learning, Reinforcement Learning, Online Learning & Optimization

The University of Texas at Austin — B.S. Computer Science & Mathematics

May 2025 | GPA: 3.65

Principles of ML, Distributed Computing, Concurrency (Honors), Stochastic Processes (Grad), Real Analysis | *QuestBridge Scholar, HSF Scholar, University Honors*

Skills

Languages: Python, SQL, Go, C/C++, TypeScript, Bash | *Familiar:* Java, Rust

AI & Agents: LLM APIs (OpenAI, Anthropic, Google), LangGraph, MCP, agentic orchestration & tool design, RAG & embedding pipelines, LiteLLM, Langfuse, agent evaluation harnesses, LoRA/QLoRA fine-tuning, knowledge graphs & ontology design

ML & Data: PyTorch, scikit-learn, pandas/NumPy/SciPy, Gymnasium, Stable-Baselines3, ChromaDB/LanceDB, Postgres

Web & Infra: Django, FastAPI, React/Next.js, GCP, AWS, Docker, Kubernetes, Keycloak/OIDC | **Spoken:** English, Spanish